

FIT (Air-cooled) Series

Industrial UV nanosecond laser

- < Ultra small size
- < Stable operation at extreme temperatures (-5 °C~40 °C)
- < Real time monitoring of power
- < One-button power optimization



► Features & Benefits:

The FIT series is an all-in-one, low-power, air-cooled, nanosecond ultraviolet 355nm laser. It features a clean and tidy blue and white colour scheme and a small, compact body structure. It weighs only 5kg and has a symmetrical appearance.

The FIT Series laser is smaller than the FOTIA Series, making it more compatible with the customer's integrated optical path and easier to install and replace. The laser interface is also more compact, retaining only the 12V power supply interface and the 15-pin signal interface in the electrical interface. We apply various technologies, including but not limited to real-time power optimization technology, new liquid-cooling technology, integrated electronic control technology, new mechanical architecture technology, etc., to achieve the development and production of third-generation FIT air-cooled lasers.

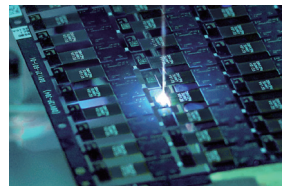
The FIT series offers a cost-effective advantage, excellent spot performance, and can reach up to 200 μ J. It has a pulse energy and power margin that averages over 30%, providing a good customer experience with convenient and simple operation, and fast integration. The laser is integrated with automatic optimization and power display, making it suitable for a wide range of applications and small spaces.



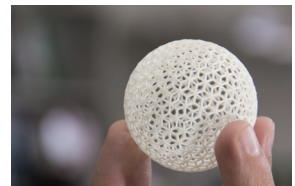
package marking



3C marking

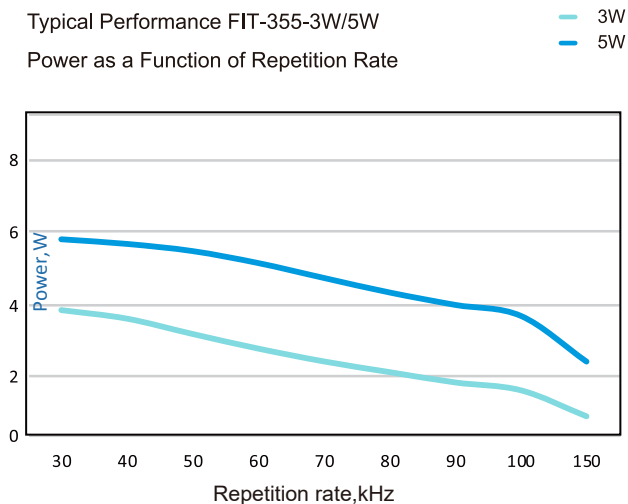


FPC/PCB marking

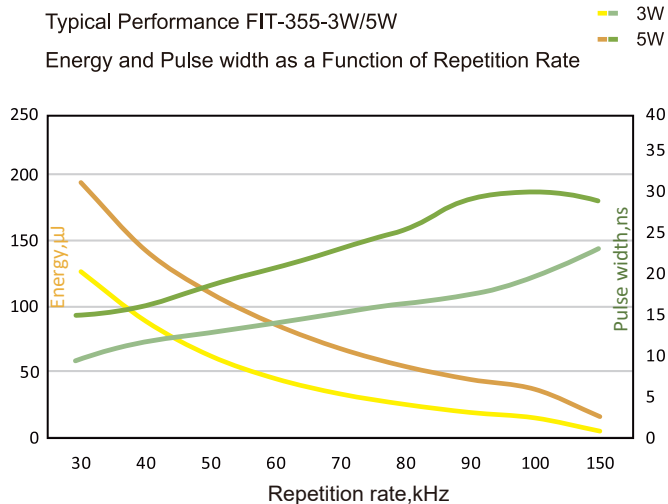


3D printing

Typical Performance FIT-355-3W/5W
Power as a Function of Repetition Rate



Typical Performance FIT-355-3W/5W
Energy and Pulse width as a Function of Repetition Rate



FIT 355		
Specification	3W-30K	5W-30K
Wavelength (nm)	355	
Average Power (Watts)	>3W@30KHz	>5W@30KHz
Energy (μJ)	>100	>160
Specified Repetition Rate(kHz)	30	30
Repetition Rate (kHz)	30 ~ 150	
Pulse Width (ns)	<15	
Beam Quality (M ²)	<1.2	
Beam Roundness (%)	>90	
Beam Diameter (mm)	~0.45	
Beam Divergence (mRad)	<1.5	
Point Stability (μrad/°C)	< 20	
Polarization Ratio	100:1 Linear,Horizontal	
Pulse-to-Pulse Stability (% RMS)	< 3	
Average Power Stability(% over12 hours)	< 3	
Cold Start Warm-Up (mins.)	< 20	
Standby Warm-Up (mins.)	< 10	
Operational Temperature Range (°C)	-5-40°C	
Operation Humidity Range (%)	20 to 80,Non-condensing	
Storage Temperature Range (°C)	- 20 to 50	
Storage Humidity Range (%)	20 to 80,Non-condensing	
Input Voltage (VDC)/Rated Power(W)	12/350	
Average Power (W)	150	
Communication	RS232	
Cooling	Air	
Weight (kg)	5	

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